



# Koneru Lakshmaiah Education Foundation

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Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

## DEPARTMENT OF CSE

### PROJECT ABSTRACT SUBMISSION

### COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053 A) A.Y. 2026 - 2027

| TITLE OF THE PROJECT |               |                   |
|----------------------|---------------|-------------------|
| S. No.               | University ID | Name              |
| 1                    | 2420030472    | Sri Raaga.T       |
| 2                    | 2420030551    | Naru Mokshita     |
| 3                    | 2420030745    | CH.Hemavathi      |
| Name of the Guide    |               | Chandra Shekhar.R |

### Abstract

The **Adaptive Digital Twin Platform for Intelligent Urban Safety and Emergency Response** is a real-time urban safety management system designed to improve the monitoring, analysis, and response to emergencies in urban environments. The platform creates a digital representation of an urban area and continuously updates it using incident and environmental data, enabling authorities to obtain a real-time view of potential safety threats and emergency situations.

The system follows the **MAPE-K (Monitor, Analyze, Plan, Execute – Knowledge)** framework to support adaptive decision-making. Data collected by the system is monitored and analyzed to identify incidents and changing conditions, while the planning and execution stages assist in selecting and coordinating appropriate emergency responses. A **React.js** dashboard with **Leaflet and OpenStreetMap** provides an interactive geographical view of incidents, while **FastAPI** manages backend services and **MySQL** stores system data. **WebSockets** enable real-time communication between the backend and dashboard, and **Grafana** provides monitoring and analytical visualizations. The complete system is designed to be containerized using **Docker** for reliable deployment.

The proposed platform aims to demonstrate how digital twins, real-time communication, and adaptive computing can be integrated to provide a scalable and intelligent solution for urban safety and emergency response.

Signature of the Guide

HOD

